| Question | | | Marking details | Marks available | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | AO1 | AO2 | AO3 | Total | Maths | Prac |
| 8 | *(a)* | | Angular acceleration is <u>rate of</u> (1)<br><u>Change of angular</u> velocity (1) | 2 | | | 2 | | |
| | *(b)* | (i) | Torque is rate of change of ang mom used (1)<br>Substitution of data (1)<br>Change in angular momentum = 156.4 kg m$^2$ s$^{-1}$ (1)<br>Answer = 68 Nm (1) | 1<br>1 | 1<br>1 | | 4 | 3 | |
| | | (ii) | Angular Momentum is conserved (1)<br>Adopting tuck position reduces moment of inertia (1)<br>Angular velocity increases (1)<br>Spin completed easier or more spins completed (1) | | 4 | | 4 | | |
| | | (iii) | Using $F = \frac{mv - mu}{t}$ (1)<br>$F$ = 1 140 N (1)<br>By N3 gymnast exerts equal & opposite force (1) | 1 | 1<br>1 | | 3 | 2 | |
| | *(c)* | (i) | Centre of gravity directly above [accept: below] rings (1)<br>No (net) moment (1) | 1 | 1 | | 2 | | |
| | | (ii) | 628 = 492 + 136 (1)<br>So no acceleration or no net force (1)<br>Clockwise moment = 492 $\times$ 3.1 = 1 525 [N cm] (1)<br>Anticlockwise moment = 136 $\times$ 12 = 1 632 [N cm] (1)<br>Will rotate anticlockwise (1) | | | 5 | 5 | 5 | |
| | | | **Question 8 total** | **6** | **9** | **5** | **20** | **10** | **0** |